Lecture 13

1. Null Space

Def: Let A be an $m \times n$ matrix. The nullspace of A is the solution set of the equation $A\vec{x} = \vec{0}$

$$\text{Null}(A) = \{\vec{x} \in \mathbb{R}^n \mid A\vec{x} = \vec{0}\}$$

Theorem: If A is an $m \times n$ matrix then $\text{Null}(A)$ is a subspace of $\mathbb{R}^n$

2. Column and Row Space

Def: Let A be an $m \times n$ matrix. The column space of A is the span of the columns of A

$$\text{Col}(A) = \text{span}\{\vec{a_1}, \dots, \vec{a_n}\}$$
$$= \{\vec{b} \in \mathbb{R}^m : A\vec{x} = \vec{b} \text{ for some } \vec{x} \in \mathbb{R}^n\}$$

Theorem: If A is an $m \times n$ matrix the $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$

Def: Let A be $m \times n$ matrix. The row space of A is the span of the rows of A

Theorem: If A is an $m \times n$ matrix the $\text{Row}(A)$ is a subspace of $\mathbb{R}^n$

3. The Kernel and Range of a Linear Transformation

Def: If $T: V \to W$ is a linear transformation, the kernel or null space of T is the set of all elements in V whose image under T is zero

$$\text{kernel of } T = \{ x \in V : T(x) = \vec{0} \}$$

Theorem: If $T: V \to W$ is a linear transformation then kernel of T is a subspace of V and the range of T is a subspace of W

Theorem: If $T: \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation with standard matrix A then

$$\begin{cases} \text{null}(A) = \text{kernel of } T \\ \text{Col}(A) = \text{range of } T \end{cases}$$